ORIGINAL RESEARCH ARTICLE

Short- and Long-Term Financial Distress Prediction in SMEs: A Survival Model Comparison

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ABSTRACT
Prediction of the financial distress of SMEs puts forward an effective aspect of risk management and decision-making. The main purpose of this study is the comparison of three survival prediction techniques: Cox Proportional Hazard, Axis-Based Random Survival Forest (RSF), and Accelerated Oblique Random Survival Forest using real SME data from Portugal. The latter technique is a novelty in the context of financial stress forecasting. Model performance quality metrics used to evaluate included the AUC, Brier Score, IBS, combined with cross-validation in a 10-fold manner across 1, 3, and 5-year predictions. The Cox model underperformed; hence, it has little potential in SMEs' financial distress prediction. Accelerated Oblique RSF gave very good results for short-term predictions, while for long-term predictions, good performances were witnessed with both Accelerated Oblique RSF and Axes-based RSF models.

KEYWORDS
Financial stress; SME; Cox Proportional Hazard Model; Random Survival Forest

1. Introduction

The contribution of Small and medium-sized enterprises (SMEs) to most economies is majorly through employment opportunities given and, in so doing, weighs positively on the general growth. Consequently, the effects of financial distress within this sector may go a long way impacting not only on the enterprises but on the general socio-economic fabric. Against such a realization, financial distress prediction and mitigation are very key in assisting effective risk management and enlightened decision-making.

This paper presents a comparative analysis in the field of financial distress prediction, more specifically related to Portuguese SMEs. Survival models are primary tools that allow analysts to extract knowledge from time-to-event data about the risk of financial difficulty and thus help to have proactive measures in order to avoid future crises.

The survival models under comparison in this study are the Cox Proportional Hazard model, the Axis-Based Random Survival Forest (Axes-based RSF), and the Accelerated Oblique Random Survival Forest (Accelerated Oblique RSF). The traditional Cox model has been the mainstay choice for survival analysis, while Axes-based RSF and Accelerated Oblique RSF represent burgeoning methodologies that manages to

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bring innovation to the table in a meaningful way. These advanced models can increase prediction accuracy, especially for short- and long-run financial distress forecasts.

We employ a Portuguese SMEs' real-world data to anchor our analysis within practical business realities with different quality metrics to be used for the measurement of the quality and performance of the models—that is, Area Under the Curve (AUC), Brier's score, and Integrated Brier's score (IBS). They are then subjected to strong internal tests using the 10-fold cross-validation method, and the predictions of the models are tested considering different time horizons, for example, 1, 3, and 5 years.

This research, therefore, demands the need for advanced survival models to be adopted in assessing financial distress risk among SMEs. Especially in an era of continuous change within the economic environment, choosing an appropriate model for the prediction horizon becomes important. Such findings also add to the growing body of knowledge in the domain of applications of survival modeling within the financial sector, having practical implications for risk management practitioners who seek to improve SMEs' resilience.

2. Literature Review

A high rate of business bankruptcies can be devastating for the economic and entrepreneurial ecosystem, especially in countries with a high concentration of Micro, Small and Medium Enterprises (SMEs) such as Portugal [4].

Financial stress, defined as a company's inability to fulfil its financial obligations, resulting in a higher proportion of liabilities or accumulation of losses [5], can lead to company bankruptcy. This reinforces the importance of effective predictive models in forecasting financial stress events, since few companies survive financial problems [8].

Over the last few decades, the development of methods for predicting corporate bankruptcies has evolved significantly, establishing itself as a relevant area of research in several disciplines, including business economics, computer science, operations research, finance and accounting. Driven by the severe impact that financial crises, such as 2007-2009, and global health crises, such as the COVID-19 pandemic, have had on companies of all sizes, a critical need has arisen to develop innovative methodologies for predicting corporate bankruptcies [50].

More traditionally, bankruptcy prediction methods were univariate analyses of financial ratios, which were already initiated by Beaver in 1966 showing such ratios could predict insolvencies. Then Altman, in 1968, developed the Z-Score model that used Multiple Discriminant Analysis (MDA) to calculate a probability of bankruptcy. With time, however, the multivariate normality of the variables was questioned, models such as logit [33] and probit [49] were thereafter adopted to provide a flexible probabilistic basis for insolvency prediction.

The Cox proportional hazards model [10], developed by Cox in 1972, finds widespread application in survival studies due to its flexibility. This model is utilized in survival analysis because it is non-parametric [4]. However, when the proportionality of risks assumption is violated, which happens frequently in financial data, the challenge arises. Thus, the need for more robust and adaptable models able to deal with the heterogeneity of corporate financial data becomes even more evident.

The 1990s marked the beginning of the incorporation of artificial intelligence algorithms into bankruptcy forecasting. Machine learning methods such as neural networks and genetic algorithms began to show promising results without the restrictions of traditional models. Research such as that by [3] has highlighted the substantial

improvement in the accuracy of bankruptcy predictions when compared to traditional methods. However, many of the algorithms used did not take into account the time until the event occurred, a fundamental aspect for financial risk modelling [44]. Recently, machine learning techniques have been applied to survival analysis in the financial context, as in the study of the time to bankruptcy of start-ups by [44] where the author concludes that the Multi-Task Logical Regression (MTLR) and Random Forest techniques have a superior performance (in terms of accuracy) compared to the Cox model.

Accuracy in predicting bankruptcies and financial stress events also depends heavily on the careful selection of financial ratios. Techniques such as PCA and LASSO [43] have been applied to identify critical predictors from extensive lists of ratios. In addition, the inclusion of non-financial variables, such as market valuation and corporate governance, has enriched bankruptcy prediction models.

It is argued that the bankruptcy and financial stress prediction model must fulfil two main requirements: high accuracy and interpretability [39]. As [34] explains, it is important for creditors, investors and banks that a clear interpretation of the results is a key aspect in determining whether the model is usable in the sector. Thus, the explainability of black box models is important in this context.

Numerous studies have focused on financial distress prediction, applying various datasets and methodologies. [45] used financial data from French companies (2002–2006) to predict financial distress, employing variables such as liquidity ratios, profitability ratios, and debt metrics. [14] applied data mining techniques on Chinese firms (2008–2011), using indicators like total liabilities, net profit ratios, and asset turnovers. [15] compared survival analysis and decision tree models using a dataset spanning from 1974 to 1991. [21] analyzed Taiwanese firms (2010–2016) with machine learning models, leveraging 16 financial ratios derived from standard financial statements.

[9] incorporated both financial and non-financial variables from the Taiwan Economic Journal (2012–2018) in their machine learning analysis, while [17] examined Slovak enterprises using traditional and machine learning methods. [40] focused on identifying optimal classifiers for UAE companies (2010–2017). [28] investigated SMEs from Estonia, Latvia, and Lithuania using a variety of ratios, while [32] assessed the impact of COVID-19 on Indian SMEs through stock price data (2011–2021).

[48] utilized AI models to predict distress in Chinese manufacturing firms (2016–2021), employing profitability, solvency, and asset turnover variables. More recent studies, such as [19] on Vietnamese firms and [42] on Chinese companies, have continued to refine prediction models using advanced machine learning algorithms. [47] predicted distress in Turkish SMEs using 47 financial indicators.

These studies employed a variety of machine learning algorithms, including Random Forest (e.g.[9, 48]), Decision Trees (e.g. [15], Support Vector Machines [45], Neural Networks (e.g. [14]), and XGBoost e.g.([21]). A summary of the algorithms used in these studies is provided in Table 1.

This paper exclusively considers research conducted within the research community that focuses on predicting financial distress in companies. Notably, there are existing studies employing bagged tree models such as random forest and survival random forest for predicting corporate bankruptcy [1]. It is important to note that the authors consider the terms "financial distress" and "bankruptcy" are not the same thing, since some factors may lead to financial distress in organizations without necessarily culminating in bankruptcy [36, 37].

To the best of our knowledge, this is the first study using Axis-Based Random

Table 1. Summary of Algorithms used for Financial Distress Prediction.

ResearchPaper	*Algorithms*
Vieira, et al., 2009 [45]	LR, NN with a Multilayer Perceptron (MLP), SVM and AdaBoost M1
Geng, et al., 2015 [14]	DT, NN, and SVM
Gepp and Kumar, 2015 [15]	Cox, DT, LR and DA
Huang and Yen, 201 [21]	SVM, XGB, Hybrid associative classifier with translation (HACT), Hybrid GA-fuzzy clustering, Unsupervised prediction model (DBN) and Hybrid supervised–unsupervised prediction model (DBN-SVM)
Chen and Shen, 2020 [9]	DT and RF
Gregova, et al., 2020 [17]	LR, NN, and RF
Sreedharan, et al., 2020 [40]	DT, NN, RF, SVM, Majority Voting and AdaBoost Ensemble
Malakauskas and Lakstutiene, 2021 [28]	LR, NN and RF
Munde and Mishra, 2022 [32]	DT and RF
Zhong and Wang, 2022 [48]	DT, LR, NN, RF, SVM, XGB and CatBoost
Ha, el al., 2023 [19]	DT, kNN, LR, RF, SVM and Bayesian Network
Tang and Xie, 2023 [42]	NB, LR, RF and SVM
Yusuf and Karavardar, 2023 [47]	DT, NB, kNN, LR, RF and SVM

[1]Decision Tree (DT), Logistic Regression (LR), k-Nearest Neighbor (kNN), Naïve Bayes (NB), Neural Networks (NN), Support Vector Machine (SVM), Random Forest (RF), XGBoost (XGB) and Discriminant Analysis (DA).

Survival Forest and Accelerated Oblique Random Survival Forest to predict financial distress of companies. However, these algorithms were already used in [46] to improve risk prediction of 1-year allograft loss in pediatric heart transplant patients compared to traditional statistical modeling using Cox regression, gradient boosting, axis-based random survival forests and oblique random survival forests.

3. Materials and Methods

3.1. Data Collection and Pre-processing

The dataset used in this research was procured from the SABI platform and involved data on 18,145 Portuguese SMEs operating within the country's manufacturing sector. The data refer to a 10-year time span, from 2007 to 2017, which will give a great opportunity to analyze in a dynamic perspective financial performance and possible distress revisions of the companies. Noteworthy enough, this dataset had previously been analyzed using the Cox Proportional Hazard model, as documented in the [4], although with less covariates than those considered in this analysis.

In our study, we employ several key variables to comprehensively analyze the dynamics of financial distress within Portuguese SMEs operating in the manufacturing sector, summarized in Table 2. These variables are reported in the literature (detailed below) as essential for understanding the various factors that influence the occurrence of financial stress events and the survival times of these enterprises.

Time (Years since firm creation until event) is a fundamental variable that tracks the duration from the creation of an SME to the occurrence of a financial stress event. It helps to understand how time plays a role in financial distress, enabling to assess the impact of the duration an SME has been in operation on its financial stability.

The **State** variable, which corresponds to (the *Censorship* variable in the survival analysis context), is used to indicate when a financial stress event has occurred during the observation period. Following to the research conducted by [18], our study defines

Table 2. Description of the Variables of the Dataset.

Variable	*Description*
Time	Years since firm creation until event
State (Censorship)	EBITDA is less than financial expenses for 2 consecutive years, and general liquidity (current assets over current liabilities) is less than 1 for 2 consecutive years.
Size	1 if Micro-sized firm; 2 if Small-sized firm; 3 if Medium-sized firm
Ownership Structure	1 if ownership concentration $\leq 50\%$
Activity Sector	Categorical variable controlling 309 different economic activities in the Manufacturing Sector
Exports	0 if domestic, 1 if exports to EU, 2 if exports to extra EU
EURIBOR 6m	Euro Interbank Offered Rate - key indicator of market conditions

a company as being in financial distress when certain criteria are met. Specifically, financial distress is recognized when a company's EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization) falls below its financial expenses for two consecutive years. Additionally, the company is considered to be in financial distress if its general liquidity, measured as the ratio of current assets to current liabilities, is less than 1 for two consecutive years. While [18] employed the solvency ratio (equity over total liabilities) to gauge financial distress, our use of the general liquidity ratio offers insights into a company's capacity to meet short-term obligations. When this ratio falls below 1, it means that the company's short-term debts, which encompass obligations to be settled within a year or less, surpass the value of assets with the highest liquidity, such as cash and cash equivalents and other assets that can readily be converted into cash within a year or less. This definition allows for a robust assessment of financial distress based on short-term liquidity constraints, a crucial aspect of a company's financial health.

To build the covariant **Size**, we categorize SMEs into three groups based on their size: the variable can take the value 1 for micro-sized, 2 for small-sized and 3 for medium-sized firms. This variable allows us to explore how the size of an SME affects its likelihood of experiencing financial distress. Smaller firms may face different challenges and risks compared to larger ones ([11],[13],[29]).

The **Ownership Structure** variable examines the concentration of ownership within an SME. When this variable takes on the value of 1, it signifies a high level of ownership concentration, typically 50% or more. Ownership concentration can influence the decision-making processes, financial strategies, and risk management approaches of an SME (based on [16], [29], [31], [41]). This is a proxy for the management characteristics of the company distinguishing firms owned at least in 50% by its largest shareholder from other ones .

The **Activity Sector** is a categorical variable that encompasses the diverse range of economic activities found within the Manufacturing Sector. In our study, it accounts for 309 different economic activities. This variable allows us to investigate the impact of the specific industry or sector in which an SME operates on its likelihood of experiencing financial distress as in the works of [35], [11] and [7].

We categorize SMEs based on their export behavior, creating the variable **Exports**. This variable can take on one of three values: 0 for firms that operate purely domestically, 1 for firms that export to the European Union (EU), and 2 for firms that export to regions outside of the EU. The export behavior of SMEs can have implications for their exposure to international markets, currency exchange risks, and market conditions, which may influence their financial stability ([35], [7]).

The variable **EURIBOR 6m** serves as a continuous variable in our analysis. It represents a key indicator of market conditions, specifically the interest rate at which

European banks lend to one another. This variable allows us to consider the broader economic context and its influence on SMEs' financial health [12]. Changes in **EURIBOR 6m** can reflect shifts in borrowing costs and financial strategies, which may impact the likelihood of financial distress.

All these variables together provide an overview of the survival times and occurrences of events related to financial distress in the life course of SMEs operating in the Portuguese manufacturing sector. They yield information on how time, firm characteristics, industry context, export behavior, and market conditions contribute to shaping the dynamics of financial distress in this key sector of the economy. Variable distribution is summarized in Table 3.

Table 3. Summary Table

	N	%
State		
0	10257	56.53
1 (= Financial Stress)	7888	43.47
Ownership Structure		
$< 50\%$	15309	84.37
$\geq 50\%$	2836	15.63
Size		
Micro	10868	59.9
Small	5580	30.75
Medium	1697	9.35
Exports		
Domestic	11346	62.53
European Union	3010	16.59
Outside European Union	3789	20.88
EURIBOR 6M Mean (sd)	1.356 (2.16)	

Data preprocessing was undertaken to ensure the dataset's suitability for survival analysis. The preprocessing steps included handling missing values, outliers, and ensuring data consistency.

3.2. Survival Models

One of the important areas in statistics, survival analysis caters to modeling time-to-event data when the "event" or response of interest can be very general, including, among others, failures, deaths, or in the context of this paper, financial distress. The underlying idea to preserve in survival analysis is the time to events of interest. It is primarily oriented to describe the distribution of survival times and to discover the predictors which determine those. In this paper, we aim to look into the issue of survival in the context of smaller companies present in the manufacturing sub-section in its struggle to predict time elapsed until financial distress has taken place.

A central concept in survival analysis is the hazard function, often denoted as $h(t)$, which represents the instantaneous rate of event occurrence at a specific time t, given that the subject (firms in our specific case) has survived up to that time. This concept is pivotal for understanding the dynamic nature of survival data.

Mathematically, the hazard function can be defined as:

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t | T \geq t)}{\Delta t} \tag{1}$$

Where:

T is the time to event (survival time). $P(t \leq T < t + \Delta t | T \geq t)$ is the conditional

probability that an event occurs in the interval $[t, t + \Delta t)$ given that the subject has survived up to time t. Δt is an infinitesimally small time interval. The hazard function provides information into how the risk of an event changes over time.

The hazard function characterizes how the risk of experiencing an event changes over time. The greater the hazard, the greater the risk that the event will happen; the lower the hazard, the smaller the risk. More importantly, the hazard function allows one to model risks of events that are time-varying and may differ across strata or different groups of subjects.

In most of the cases, in survival data analysis, some subjects have not yet experienced the event of interest at the end of the observation period. It is what may be termed right censoring—one of the distinguishing features of survival analysis. Right censoring occurs when only part of the information on times to events is available for some subjects. This is usually due to the consequence of a long-term study or observational data. Handling of right censoring is important to be able to include people still in observation and who will probably suffer the event in the future.

In the context of right-censored survival data, the observed dataset is represented as (T, δ), where δ serves as the censoring indicator. The observed time T is defined as the minimum value between the true (though potentially unobserved) survival event time, denoted as T_0, and the true (likewise potentially unobserved) censoring time, denoted as C_0. Consequently, we have $T = min(T_0, C_0)$, where it's important to note that the actual event time may remain unobserved. The censoring indicator, δ, is defined as $\delta = \mathbb{I}\{T_0 \leq C_0\}$. If δ equals 1, it indicates that an event has occurred, meaning that we observe the true event time, $T = T_0$. Conversely, when δ equals 0, the observation is censored, and we solely observe the censoring time, $T = C_0$. Thus, in such cases, we can ascertain that the subject has survived up to time C_0, but the exact time of the event remains unobserved.

In the next sections we briefly explain the Cox Proportional Hazard Model (CPHM), the Axis-based Random Survival Forest (RSF) and the Oblique Random Survival Forest (ORSF).

3.2.1. Cox Proportional Hazard Model (CPHM)

The Cox Proportional Hazard model, a well-established survival model proposed by Cox [10], was employed as a benchmark for comparison.

In the Cox Proportional Hazard model, the hazard function is expressed as a function of n covariates, x_{in}, and a baseline hazard $h_0(t)$, as follows:

$$h_i(t) = h_0(t) \cdot \exp\left(\beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik}\right) \tag{2}$$

and it follows the proportional hazards assumption, not time dependent expressed as:

$$\frac{h_i(t)}{h_j(t)} = \frac{h_0(t) \cdot \exp(\beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik})}{h_0(t) \cdot \exp(\beta_1 x_{j1} + \beta_2 x_{j2} + \ldots + \beta_k x_{jk})} \tag{3}$$

3.2.2. Axis-Based Random Survival Forest (RSF)

The Random Survival Forest (RSF) is an extension of random forest method proposed by [6], for the analysis of right-censored survival data. It is an ensemble learning tech-

nique that extends the concept of random forests to time-to-event data. This model leverages decision tree ensembles to account for complex, non-linear relationships between predictor variables and the event of interest [25].

As described by [30] the algorithm for constructing an ensemble of trees to predict survival outcomes based on a given set of covariates, can be summarized as follows:

(1) Generate B (ntree) subsamples from the original dataset, each of a specified sample size, drawn with or without replacement.
(2) Build an individual survival tree for each subsample, where b ranges from 1 to B.
    - (a) At each node within a tree, a subset of predictor variables is considered as potential candidates for splitting (mtry).
    - (b) The node split is selected from the candidate variables in a way that maximizes the survival difference between the resulting daughter nodes, guided by a chosen split criterion (splitrule) and up to a predefined number of split points (nsplit).
    - (c) This process is applied recursively on each daughter node until a stopping criterion is met, often based on the number of unique cases within each terminal node (nodesize).
(3) Compute the cumulative hazard function for each tree by utilizing the terminal nodes. Subsequently, aggregate the information derived from the B survival trees to produce a predictive ensemble that quantifies risk.

In Axes-based RSF, the survival probability for an individual at time t is calculated as a weighted combination of survival probabilities along multiple axes. The equation for the survival probability $S(t)$ is expressed as follows:

$$S(t) = \prod_{j=1}^{m} S_j(t)^{\alpha_j} \tag{4}$$

Where:
$S(t)$ represents the overall survival probability at time t, m is the number of axes, $S_j(t)$ is the survival probability along the $j-th$ axis at time t, and α_j is the weight associated with the $j-th$ axis.

As the R software package *randomForestSRC* [22] allows to choose different split criteria (splitrule), we compared in the present analysis the following three options:

(1) splitrule="logrank" (default splitrule): log-rank splitting [27, 38] The log-rank test is used to compare the survival curves of two or more groups to determine if there is a statistically significant difference in survival times. In the context of Random Survival Forests (RSF), this rule selects the variable and split point that maximizes the dissimilarity in survival between the child nodes when the node is split. It aims to maximize the statistical difference in survival curves.
(2) splitrule="logrankscore": log-rank score splitting [20]. The "logrank score" is a variation of the log-rank test that focuses on efficiency. Instead of directly comparing survival curves, it uses a scoring mechanism that quantifies the difference in survival between groups. In the context of RSF, the log-rank score splitting rule selects variables and split points that maximize the log-rank score, indicating

the largest dissimilarity in survival times between child nodes.

(3) splitrule="bs.gradient": gradient-based (global non-quantile) brier score splitting. The Brier score measures the mean squared difference between predicted probabilities of survival and the actual outcomes. The gradient-based splitting rule in RSF optimizes the Brier score by selecting variables and split points that minimize prediction errors. The time horizon used for calculating the Brier score is typically set to the 90th percentile of the observed event times, but you can customize it with the "prob" option.

3.2.3. Oblique Random Survival Forest (Oblique RSF)

Oblique Random Survival Forest (Oblique RSF) is an extension of the Random Survival Forest (RSF) method that combines survival analysis with the principles of random forests. The primary difference lies in the way variables are selected for splitting nodes in the trees. While traditional RSF uses a "parallel" splitting rule, Oblique RSF employs an "oblique" or tilted splitting rule, allowing for more complex relationships between variables.

In Oblique RSF, the splitting rule is defined as:

$$X_j - \beta \cdot X_k \leq c \tag{5}$$

Where: X_j represents one predictor variable. X_k represents another predictor variable. β is a coefficient representing the tilt of the split and c is a constant threshold.

The goal is to find the best combination of (X_j, X_k, β, c) that maximizes the separation between the survival curves, much like in traditional RSF.

The splitting rule allows for non-linear relationships between variables and can capture more complex interactions, making Oblique RSF a powerful tool for survival analysis when the relationships between covariates are not purely parallel.

In 2019 Jaeger et al. [24] developed the obliqueRSF package [23] for R software that implements this algorithm. However, as the implementation was rather slow, in 2022 they developed the oarsf package [26] that implemented the acceletated oblique RSF, which is computationally faster.

To accelerate the oblique RSF, they propose to identify linear combinations of predictor variables in non-leaf nodes by applying Newton-Raphson scoring to the partial likelihood function of the Cox regression model as follows:

$$L(\beta) = \prod_{i=1}^{m} \frac{e^{x_{j(i)}^T \beta}}{\sum_{j \in R_i} e^{x_j^T \beta}} \tag{6}$$

where R_i is the set of indices, j, with $T_j \geq t_i$ (i.e., those still at risk at time t_i), and $j(i)$ is the index of the observation for which an event occurred at time t_i.

The Accelerated Oblique Random Survival Forest employs decision tree ensembles but incorporates an accelerated failure time framework, allowing for a more flexible and robust modeling of survival data [25, 26].

In this analysis, three distinct methods for identifying linear combinations of predictors within the framework of Accelerated Oblique RSF, available at the *oarsf* package were employed. These methods were chosen based on their specific characteristics and applications, and can be summarize as follows:

Table 4. Description of the used Models.

Model	*Description*
CPHM	Cox Proportional Hazard Model
RSF_LR	Axed-based Random Survival Forest with log-rank node splitting rule
RSF_LRS	Axed-based Random Survival Forest with log-rank score node splitting rule
RSF_BS	Axed-based Random Survival Forest with Brier Score Gradient node splitting rule
AORSF	Accelerated Oblique Random Survival Forest
ORSF_CPH	Oblique Random Survival Forest
ORSF_NET	Oblique Random Survival Forest

- orsf_control_fast(): This method utilizes a single iteration of the Newton-Raphson scoring algorithm. By focusing on a single iteration, it offers a relatively faster approach for identifying linear combinations of predictors. While speed is a key advantage, it may be more suitable for situations where computational efficiency is a priority.
- orsf_control_cph(): In contrast, the orsf_control_cph() method employs the Newton-Raphson scoring algorithm until a specified convergence criteria are met. This approach ensures that the method continues to iterate until it reaches convergence. The use of the Cox Proportional Hazards (CPH) model, a common choice in survival analysis, with this control setting may provide more accurate results when identifying linear predictor combinations. It is particularly valuable when precision and model convergence are of paramount importance.
- orsf_control_net(df_target = 3): Inspired by Jaeger (2019), this method allows AORSF to incorporate the glmnet algorithm. The glmnet algorithm provides a more sophisticated approach to identifying linear combinations of predictors. It introduces regularization, allowing the model to account for variable selection and potentially enhance predictive performance. This method is particularly useful when the goal is to strike a balance between model complexity, variable selection, and prediction accuracy.

The selection of the appropriate method depends on the specific research objectives, computational resources, and the desired balance between speed and accuracy in identifying linear predictor combinations within the AORSF framework.

In synthesis, the models to be compared are presented in table 4.

3.3. Quality Assessment Metrics

To evaluate the predictive performance of each survival model, a set of quality assessment metrics was used, including:

Area Under the Curve (AUC): AUC measures the model's ability to discriminate between distressed and non-distressed SMEs, with higher values indicating better discrimination. AUC is computed by plotting the Receiver Operating Characteristic (ROC) curve. The ROC curve represents the true positive rate (sensitivity) on the y-axis and the false positive rate on the x-axis as the probability threshold varies. The AUC is the area under this curve, which ranges from 0 to 1, with higher values indicating better discrimination. Mathematically, AUC can be defined as the integral of the ROC curve.

Brier Score: The Brier Score is a metric used to assess the accuracy of predicted probabilities in survival analysis. It quantifies the mean squared difference between the predicted probabilities and the actual outcomes.

For each subject, let Y be the binary outcome (1 for distressed, 0 for non-distressed), and P be the predicted probability of the event occurring. The Brier Score is calculated as the mean squared difference between Y and P across all subjects:

$$BS = \frac{1}{N} \sum_{i=1}^{N} (Y_i - P_i)^2 \tag{7}$$

where N is the number of subjects. Lower values of the Brier Score indicate superior accuracy of the model's probability prediction, i.e., better-calibrated predictions.

Integrated Brier Score (IBS): IBS or or IPA (Integrated Prediction Accuracy), is a metric used in survival analysis to evaluate the overall model performance across different time horizons. It provides a comprehensive assessment of the model's predictive accuracy over time.

IBS is often defined as the time-weighted average of Brier Scores at different time points:

$$IBS = \int_0^T \text{Brier Score}(t) \cdot w(t)\, dt \tag{8}$$

where T is the maximum time horizon, Brier Score(t) is the Brier Score at time t, and $w(t)$ is a weighting function that determines the importance of different time points in the evaluation.

3.4. Implementation

The cross-validation technique used to check the models' robustness and generalization capability was k-fold cross-validation. In this case, a 10-fold cross-validation entailed data partitioning into ten separate sets each, out of which any one passed in turn formed a test set, and nine others used for model training. The process is repeated ten times so that at some point, each of the subsets gets to become the test set.

One of the principal utilities of k-fold cross-validation is to escape overfitting, which generally occurs during the performance measurement of machine learning models and provides a guess on their suitability with respect to unknown data. This will thus aid in properly assessing the predictive ability of each model with time dummies, more so to the 1-year, 3-year, and 5-year predictions of financial distress.

This will be essential in order to test most rigorously the effectiveness of models in describing the financial distress of Portuguese SMEs in predictions by survival models at a completely unbiased level, after which performances can be compared robustly and accordingly an effective contribution can be made to socio-economic planning of the manufacturing sector in that region.

4. Results

In this section, we present the results of our analysis, which involved comparing seven different survival analysis models using a 10-fold cross-validation for 1, 3, and 5-year predictions. The primary metrics of interest were the Area Under the Curve (AUC) and the Brier Score, along with their associated confidence intervals and p-values reported in A. Additionaly, the IPA Score is also compared. We also examined the performance differences between models and provided descriptions of the models for context.

Figures 1 and 2 summarize the performance, through values of AUC and Brier Score, of the seven survival analysis models for predicting events at different time horizons. The lower and upper bounds of the 95% confidence intervals provide an understanding of the precision of our estimates.

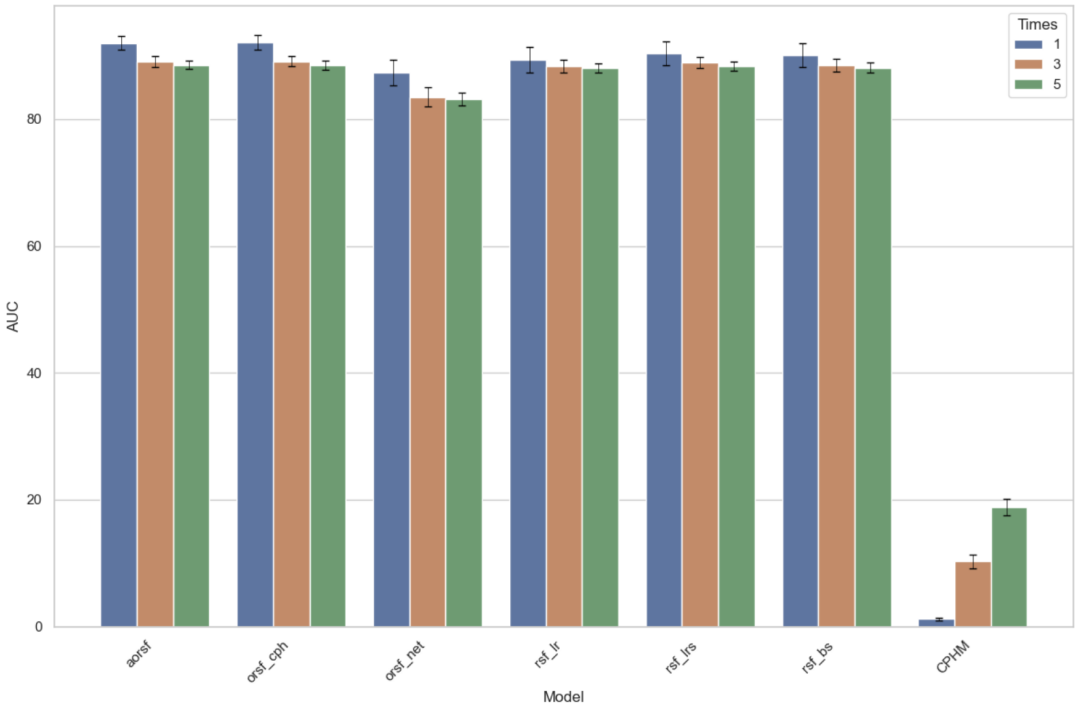

Figure 1. AUC results for 1, 3, and 5-year Prediction (with 95% confidence intervals bar errors).

In the context of 1-year predictions, we observe variations in model performance. Model aorsf and Model orsf_cph stand out with significant differences in AUC compared to the other models. Model aorsf achieves the highest AUC of 92 and a Brier Score of 0.9, while Model orsf_cph follows closely with an AUC of 92.1 and a Brier Score of 0.9. These models demonstrate their superiority in short-term event prediction.

Models rsf_lr, rsf_lrs, and rsf_bs display relatively similar AUC values, ranging between 88.4 and 90.4, with corresponding Brier Scores of 0.9, indicating a balance between discrimination and calibration. While they do not significantly outperform the top two models, their consistency indicates reliability in short-term predictions.

Model CPHM exhibits a substantially lower AUC of 1.2 for 1-year predictions, but its Brier Score is 84.2, suggesting a potential advantage in calibration, although with limited discrimination.

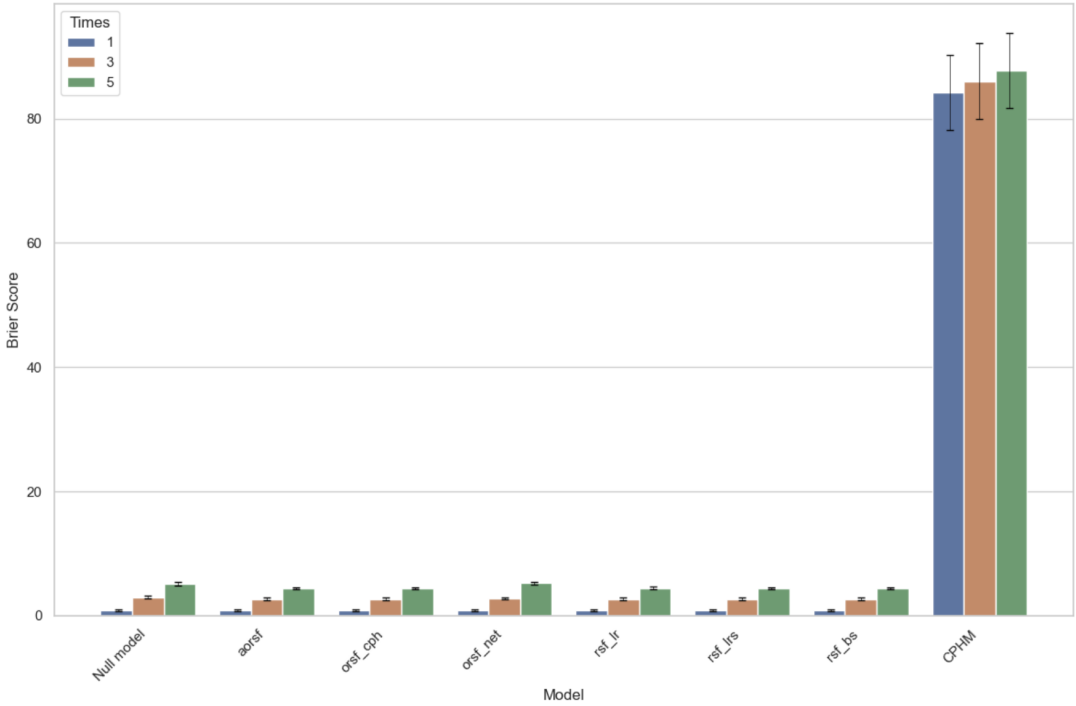

Figure 2. Brier results for 1, 3, and 5-year Prediction (with 95% confidence intervals bar errors).

In contrast, the comparisons for long-term predictions (3 and 5 years) reveal distinct shifts in model rankings.

Model aorsf, Model orsf_cph, and Model orsf_net maintain their dominance across all time horizons. While their AUC values differ, their Brier Scores remain close. Model aorsf demonstrates consistent performance with Brier Scores of 0.9, 0.9, and 0.9 for 1, 3, and 5-year predictions, respectively. Model orsf_cph exhibits similar consistency in Brier Scores, maintaining the trend of short-term superiority.

Models rsf_lr, rsf_lrs, and rsf_bs also display relative consistency in their performance, with similar Brier Scores. While they do not surpass the top models in AUC, their stability across different timeframes suggests their suitability for long-term event prediction.

Model CPHM showcases a significant improvement in both AUC and Brier Score as the prediction horizon extends. Its AUC increases from 1.2 for 1-year predictions to 10.3 for 5-year predictions, and its Brier Score drops from 84.2 to 18.8. This indicates that Model CPHM excels in long-term forecasting, exhibiting both improved discrimination and calibration, but remains the one with lower prediction accuracy compared to Random Survival Forests, axes or obliqued based.

The IPA score, presented in Figure 3, measures the improvement in predictive accuracy of a model compared to a baseline (null) model. A score of zero suggests no improvement over the baseline.

For 1-year predictions, models such as aorsf and orsf_cph demonstrated a slight enhancement in predictive accuracy compared to the null model, with IPA scores of 4.2 and 4.3, respectively. Additionally, random survival forest variations (rsf_lr, rsf_lrs, rsf_bs) showed some improvement over the null model, though to a lesser extent, with

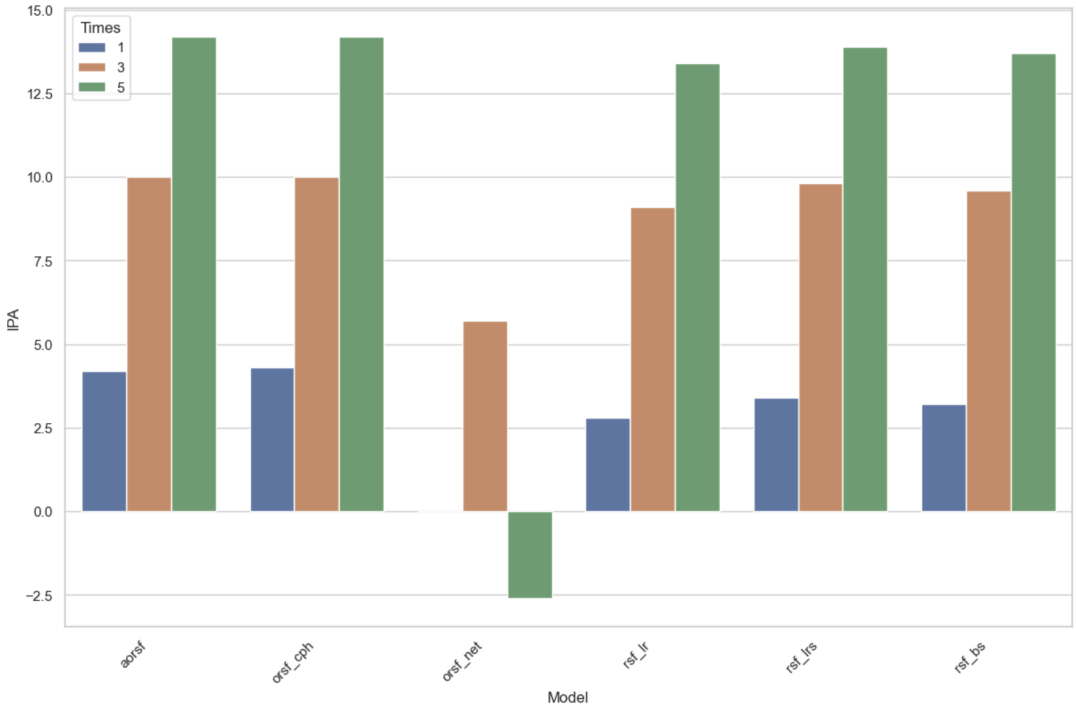

Figure 3. IBS results for 1, 3, and 5-year Prediction

IPA scores ranging from 2.8 to 3.4. Conversely, orsf_net displayed no improvement (IPA score of 0). While not presented in the figure, due to its extreme values compared to other models (which would make reading the graph impractical), the Cox Proportional Hazard Model (CPHM) exhibited a lower performance with an IPA score of -8814.1, indicating substantial deviation or issues compared to the null model.

Moving to 3-year predictions, both aorsf and orsf_cph models exhibited substantial improvements over the null model, with high IPA scores of 10. Similarly, the random survival forest variations displayed improvements over the null model, ranging from IPA scores of 9.1 to 9.8. Meanwhile, orsf_net showed moderate improvement (IPA score of 5.7). Conversely, the CPHM continued to exhibit a notably lower performance with an IPA score of -2761.

For 5-year predictions, aorsf and orsf_cph models continued to demonstrate an improvement over the null model, with IPA scores of 14.2. The random survival forest variations also displayed improvements over the null model, though at a lower level, with IPA scores ranging from 13.4 to 13.9. However, orsf_net showed a slight negative improvement (IPA score of -2.6), indicating a decrement in predictive accuracy. The CPHM still underperformed significantly compared to the null model with an IPA score of -1614.3 across this time interval.

In summary, the aorsf, orsf_cph models, and the random survival forest variations consistently outperformed the null model across different time intervals. However, the Cox Proportional Hazard Model consistently exhibited notably poorer predictive accuracy compared to the null model, suggesting potential limitations or issues in its performance.

Thus, based on the IPA scores presented in the table, aorsf and orsf_cph models appear to be the most favorable choices across different time intervals for predicting

financial distress in small and medium-sized enterprises (SMEs). These models consistently showed higher predictive accuracy compared to both the null model and other model variations evaluated in this analysis.

5. Conceptual Comparison in Terms of Interpretability

It is important to also compare these models in terms of interpretability for econometric purposes, such as how each model communicates the intensity and orientation of the covariate effect.

A classical and universally applied method in survival analysis, particularly in the situation of time-dependent covariates, is the Cox Proportional Hazard Model. It gives the hazard ratios, which specify the relative contribution of each of the covariates to the risk or hazard of experiencing an event, like financial distress. The hazard ratios are pointing out what happens to the hazard rate for a one-unit change in the predictor's variables. It is a great strength of this model because it can relate direction and size effects of the covariates on survival times.

It assumes proportional hazards, which may sometimes not be true, and it could become less efficient for fitting complex interactions between the predictors. In this specific analysis, the test-statistic calculates p-values, based on Schoenfeld residuals, for the predictors, as seen in Table 5. The p-values, which are summarized by the test statistic, measure the relation between the scaled Schoenfeld residuals and time. The theoretical assumptions hold if the obtained result of the test statistic is non-significant for each of the predictors, a case which explains the really low accuracy results.

Table 5. Results of Cox proportional hazards Assumption Test.

Variable	χ^2	df	p
Size	53.6	1	2.5×10^{-13}
Exports	100.0	1	$< 2 \times 10^{-16}$
Ownership Structure	122.7	1	$< 2 \times 10^{-16}$
Activity Sector	14.8	1	0.00012
EURIBOR 6M	18.8	1	1.4×10^{-5}
GLOBAL	276.0	5	$< 2 \times 10^{-16}$

In contrast, the Axis-Based Random Survival Forest method extends, on one hand, the flexibility of the conventional RSF by the construction of a low-dimensional representation of the feature space so as to render it interpretable. This could give exposure to the dominant directions or combinations of predictors in the high-dimensional space affecting the forecast of financial distress. The representation that results is very efficient in dimension reduction and registration of the interactions but can be extremely ambivalent without the support of further analysis or visualization carried out for each of the transformed axes concretely.

With respect to the Oblique Random Survival Forest, the model allows for split node creation that is not constrained to only align with the axes of features. It captures, with quite some efficiency, the complex interacting more advantaged. As with the traditional RSF, it provides variable importance measures accommodating intricate interactions. This approach is probably able to handle the nonlinear and interacting effects of the covariates on the prediction of financial distress. However, when the model's strength lies in capturing complex relationships, complexity is brought into the interpretations of results because of the presence of non-orthogonal splits.

On one hand, although the Cox proportional hazard model directly provides haz-

ard ratios for the effects of the covariates, the model has been considered somewhat limited due to assumptions that are quite stringent in relation to the proportionality of hazards and simple interactions. On the other hand, Axis-Based and Oblique Random Survival Forests aim to improve interpretability through dimensionality reduction and allow for complex interactions. However, their transformed representations or non-orthogonal splits may not be easily interpretable without additional analysis or visualization techniques.

Random Survival Forest models are inherently complex due to ensemble techniques and handling high-dimensional data; however, they offer interpretability through multiple avenues. Variable importance in the algorithms is what makes them interpretable. This is achieved from the ability to show the number of cases that would have misled the system that it dealt with if a variable was not present. The RSF models can thus be in a position of explaining which predictors affect most the resultant outcome of survival, say, from financial distress amongst SMEs. From then, one can zero in on those parameters that besides minor adjustment have a huge effect in impacting the predictions.

For instance, results of variable importance obtained by fitting aorsf are shown in Table 6. These results reveal that the highest-ranking predictor in explaining the variability of the outcome variable is the ownership structure predictor, with an importance score of 0.1452 as compared to the other predictors in the list. It can be seen that EURIBOR 6M has an importance of 0.0843, the second highest regarding the contribution relatively in explaining the variability to the outcome, though toward the variable itself slightly smaller than the last. The remaining covariates have, in order of importance, an importance score of 0.0813 for Exports, 0.0775 for Size, and 0.0573 for Activity Sector. These are relative importance scores that suggest the following ranking in terms of explaining the variation of the outcome variable: Exports is more important compared to Size, while Activity Sector has very minimal effect compared to the rest.

Table 6. Variable Importance Scores.

Predictor	Importance Score
Ownership Structure	0.14522115
EURIBOR 6M	0.08425387
Exports	0.08128533
Size	0.07750371
Activity Sector	0.05725473

On the other hand, RSF models can generate survival prediction curves which present the estimated survival probabilities over time for individual observations or subgroups of individual observations conditional on different groups of covariate values. These are graphical tools that analysts can use to highlight and compare various predictors' effects on survival probabilities.

The node splitting rules in the RSF trees provide even more insight as to how the model divides the data according to various values of predictors. These rules can sometimes be quite complex, but will eventually help in isolating the most influential variables that result in distinct survival groups.

RSF models produce conditional effects, defined generally as changes in the predicted survival probabilities in relation to varying values of some particular covariates. Insight of this nature gives a clear view with respect to the direction and the magnitude of effect that such predictors have on financial stress prediction.

Moreover, partial dependence plots are used to display the marginal effects of individual variables toward predicting survival probabilities by holding all other variables

at some fixed value. It helps in explaining the relationship that exists between predictors and the likelihood of financial distress and hence proves helpful for interpretation.

Moreover, the RSF models have good predictive abilities in capturing interactions among predictors that eventually affect survival outcomes. To be sure, the assessment of these interactions is very informative as to how combinations of predictors jointly drive or influence the likelihood of experiencing financial distress.

6. Discussion and Conclusion

Accurate prediction and mitigation of financial distress are extremely important for the sustained growth and success of SMEs within the manufacturing sector in such a rapidly changing environment. This study is set to help in streamlining and improving models predicting financial distress for SMEs.

Actually, in this analysis, which covers the period from 2007 to 2017 with detailed financial information pertaining to the manufacturing industries from 18,145 Portuguese SMEs, the different factors at play with respect to SME financial distress are focused on.

This research work, however, took a different turn by properly analyzing seven various models of machine learning in financial distress prediction. It contained models like the Cox Proportional Hazard Model, Random Survival Forest with the log-rank node-splitting rule, Random Survival Forest with the Log-rank Score node-splitting rule, Random Survival Forest with the Brier Score Gradient node-splitting rule, Accelerated Oblique Random Survival Forest, Oblique Random Survival Forest, and Oblique Random Survival Forest with the net data frame target.

The Cox Proportional Hazard Model (CPHM) is a well-established model for survival analysis, while the RSF models use different node splitting rules in Random Survival Forests. AORSF and ORSF_CPH are specialized models with unique features and controls, offering flexibility in modeling.

The choice between the Cox Proportional Hazard model, Random Survival Forest, Axis-Based Trees, and Oblique Trees would depend on the nature of the survival data. While a Cox model is very useful in many ways, it assumes proportional hazards; therefore, should this not be the case, then random survival forest and in particular oblique trees will give a much more flexible way to accommodate complex and non-proportional survival data patterns. In the Random Survival Forest framework, Axis-Based versus Oblique Trees should be chosen on the basis of the structure of data and more importantly due to the need of capturing linear or nonlinear relationships among predictor variables.

Next, we evaluated these models' predictive performance concerning the area under the receiver operating characteristic curve and the Brier score through a rigorous 10-fold cross-validation process. It gave different results for the deep analysis of patterns about the models' capacity to predict financial stress accurately, thus giving meaning to the critical role machine learning techniques play in enhancing prediction accuracy for the varied time frames.

This means different performances for several survival-analytic models, some of them showing remarkable improvements compared to the baseline models. Model choice would depend on context and on the goals of any given survival analysis work, along with the time horizon of interest.

Models AORSF and RSF_BS seemed to bring about both the highest AUC and the lowest Brier score when predicting a year ahead. The latter models showed outstanding

predictive power in capturing the events of financial distress within the manufacturing sector and might be informative for stakeholders to engage in early intervention and mitigation efforts.

However, as we extended the prediction horizon to three or five years, we saw that RSF_LR and RSF_LRS began to outperform other models. Their rising AUC values and reduced Brier scores prima facie indicate their proficiency at modeling complexities of the financial distress events happening in the medium to long term.

These results suggest that model performance varies depending on the prediction time horizon. While certain models perform well in short-term forecasting, others show their real prowess in event forecasting at longer terms. Therefore, researchers and practitioners should ensure they consider the specific time horizon of interest when choosing the most appropriate survival analysis model for particular applications.

Comparisons of machine learning models for the prediction of financial distress will allow the stakeholders to decide on models adapted to their individual time horizons and objectives. Such exploitation of predictive strengths by these models will help in the effective design of risk management, financial support, and sustainability plans.

Furthermore, considering that different models work best in different temporal contexts, the stakeholders will be able to come up with a dynamic risk management approach which changes boundaries with respect to the time horizon and market conditions. This is considered straightforward to respond to the evolving challenges of SMEs within the manufacturing sector.

One major limitation of this study is that it was totally dependent on real data, and generalization to all scenarios is restricted due to the known type of mechanism underlying the data.

In this respect, future research in the area should simulate data in such a context, as done by [2] in the health area. Also, integration with hybrid models that integrate the strengths of various algorithms should be done. Such models can provide even more robust predictions, matching the complex dynamics of SMEs in the manufacturing sector.

Moreover, the introduction of macroeconomic and industrial factors can improve the predictive power of the models even more. Keeping pace with these external drivers of change is paramount as financial markets continue to evolve.

One of the contributions of this research, therefore, is to add to understanding in this area pertaining to the context of financial distress prediction for Portuguese SMEs in the manufacturing industry. If one thing is foretold by the performance assessment and prediction of models explored, it is that machine learning approaches will assume a relevant role in figuring out the future for SME risk management.

7. Acknowledgments

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Appendix A. T-tests for Metrics Comparison

Table A1. Significance of Mean Differences Comparison T-tests for AUC values obtained by 10-fold (1 year).

Model	Reference	Delta.AUC	Lower	Upper	p
aorsf_lc	aorsf	0.1	-0.1	0.3	5.00E-01
aorsf_lcpc	aorsf	-4.6	-6.5	-2.8	1.00E-06
rfsrc_lr	aorsf	-2.6	-4.0	-1.2	3.00E-04
rfsrc_lrsc	aorsf	-1.6	-2.9	-0.4	1.00E-02
rfsrc_bs	aorsf	-1.9	-3.2	-0.6	4.00E-03
cox	aorsf	-90.8	-91.8	-89.8	0.00E+00
aorsf_lcpc	aorsf_lc	-4.7	-6.5	-2.9	6.00E-07
rfsrc_lr	aorsf_lc	-2.6	-4.0	-1.3	2.00E-04
rfsrc_lrsc	aorsf_lc	-1.7	-3.0	-0.4	8.00E-03
rfsrc_bs	aorsf_lc	-2.0	-3.3	-0.7	2.00E-03
cox	aorsf_lc	-90.8	-91.9	-89.8	0.00E+00
rfsrc_lr	aorsf_lcpc	2.1	-0.1	4.3	7.00E-02
rfsrc_lrsc	aorsf_lcpc	3.0	0.9	5.1	5.00E-03
rfsrc_bs	aorsf_lcpc	2.7	0.5	4.9	1.00E-02
cox	aorsf_lcpc	-86.1	-88.2	-84.1	0.00E+00
rfsrc_lrsc	rfsrc_lr	0.9	0.0	1.8	5.00E-02
rfsrc_bs	rfsrc_lr	0.7	-0.2	1.5	1.00E-01
cox	rfsrc_lr	-88.2	-90.2	-86.3	0.00E+00
rfsrc_bs	rfsrc_lrsc	-0.3	-1.0	0.5	5.00E-01
cox	rfsrc_lrsc	-89.1	-90.9	-87.4	0.00E+00
cox	rfsrc_bs	-88.9	-90.7	-87.1	0.00E+00

Table A2. Significance of Mean Differences Comparison T-tests for AUC values obtained by 10-fold (3 years).

Model	Reference	Delta.AUC	Lower	Upper	p
aorsf_lc	aorsf	0.1	-0.1	0.2	4.00E-01
aorsf_lcpc	aorsf	-5.6	-7.0	-4.1	9.00E-14
rfsrc_lr	aorsf	-0.7	-1.2	-0.2	5.00E-03
rfsrc_lrsc	aorsf	-0.2	-0.7	0.3	5.00E-01
rfsrc_bs	aorsf	-0.6	-1.1	-0.1	3.00E-02
cox	aorsf	-78.8	-79.9	-77.7	0.00E+00
aorsf_lcpc	aorsf_lc	-5.6	-7.1	-4.1	6.00E-14
rfsrc_lr	aorsf_lc	-0.8	-1.3	-0.3	2.00E-03
rfsrc_lrsc	aorsf_lc	-0.2	-0.7	0.3	4.00E-01
rfsrc_bs	aorsf_lc	-0.6	-1.1	-0.1	2.00E-02
cox	aorsf_lc	-78.9	-80.0	-77.7	0.00E+00
rfsrc_lr	aorsf_lcpc	4.8	3.3	6.4	4.00E-10
rfsrc_lrsc	aorsf_lcpc	5.4	3.9	6.9	3.00E-12
rfsrc_bs	aorsf_lcpc	5.0	3.5	6.5	1.00E-10
cox	aorsf_lcpc	-73.2	-74.9	-71.5	0.00E+00
rfsrc_lrsc	rfsrc_lr	0.5	0.3	0.8	7.00E-05
rfsrc_bs	rfsrc_lr	0.1	-0.0	0.3	6.00E-02
cox	rfsrc_lr	-78.1	-79.3	-76.9	0.00E+00
rfsrc_bs	rfsrc_lrsc	-0.4	-0.6	-0.1	3.00E-03
cox	rfsrc_lrsc	-78.6	-79.8	-77.4	0.00E+00
cox	rfsrc_bs	-78.2	-79.4	-77.0	0.00E+00

Table A3. Significance of Mean Differences Comparison T-tests for AUC values obtained by 10-fold (5 years).

Model	Reference	Delta.AUC	Lower	Upper	p
aorsf_lc	aorsf	-0.0	-0.1	0.1	4.00E-01
aorsf_lcpc	aorsf	-5.4	-6.2	-4.5	5.00E-36
rfsrc_lr	aorsf	-0.4	-0.8	-0.1	8.00E-03
rfsrc_lrsc	aorsf	-0.3	-0.6	0.1	2.00E-01
rfsrc_bs	aorsf	-0.4	-0.7	-0.1	2.00E-02
cox	aorsf	-69.7	-71.0	-68.5	0.00E+00
aorsf_lcpc	aorsf_lc	-5.3	-6.2	-4.5	2.00E-35
rfsrc_lr	aorsf_lc	-0.4	-0.7	-0.1	1.00E-02
rfsrc_lrsc	aorsf_lc	-0.2	-0.6	0.1	2.00E-01
rfsrc_bs	aorsf_lc	-0.4	-0.7	-0.0	0.00E+00
cox	aorsf_lc	-69.7	-70.9	-68.4	0.00E+00
rfsrc_lr	aorsf_lcpc	4.9	4.0	5.8	1.00E-26
rfsrc_lrsc	aorsf_lcpc	5.1	4.2	6.0	2.00E-29
rfsrc_bs	aorsf_lcpc	5.0	4.1	5.9	2.00E-27
cox	aorsf_lcpc	-64.3	-65.8	-62.9	0.00E+00
rfsrc_lrsc	rfsrc_lr	0.2	-0.0	0.4	8.00E-02
rfsrc_bs	rfsrc_lr	0.0	-0.1	0.2	4.00E-01
cox	rfsrc_lr	-69.3	-70.6	-68.0	0.00E+00
rfsrc_bs	rfsrc_lrsc	-0.1	-0.3	0.0	1.00E-01
cox	rfsrc_lrsc	-69.5	-70.7	-68.2	0.00E+00
cox	rfsrc_bs	-69.3	-70.6	-68.0	0.00E+00

Table A4. Significance of Mean Differences Comparison T-tests for Brier Score values obtained by 10-fold (1 year).

Model	Reference	Delta.Brier	Lower	Upper	p
aorsf	Null model	-0.0	-0.1	-0.0	4.643e-10
aorsf_lc	Null model	-0.0	-0.1	-0.0	2.268e-10
aorsf_lcpc	Null model	0.0	-0.0	0.0	9.847e-01
rfsrc_lr	Null model	-0.0	-0.0	-0.0	3.717e-03
rfsrc_lrsc	Null model	-0.0	-0.1	-0.0	4.449e-04
rfsrc_bs	Null model	-0.0	-0.0	-0.0	6.225e-04
cox	Null model	83.2	77.2	89.3	4.291e-159
aorsf_lc	aorsf	-0.0	-0.0	-0.0	3.688e-02
aorsf_lcpc	aorsf	0.0	0.0	0.1	3.820e-08
rfsrc_lr	aorsf	0.0	0.0	0.0	1.504e-03
rfsrc_lrsc	aorsf	0.0	-0.0	0.0	8.629e-02
rfsrc_bs	aorsf	0.0	0.0	0.0	2.114e-02
cox	aorsf	83.3	77.2	89.3	2.853e-159
aorsf_lcpc	aorsf_lc	0.0	0.0	0.1	1.697e-08
rfsrc_lr	aorsf_lc	0.0	0.0	0.0	5.640e-04
rfsrc_lrsc	aorsf_lc	0.0	0.0	0.0	4.898e-02
rfsrc_bs	aorsf_lc	0.0	0.0	0.0	9.467e-03
cox	aorsf_lc	83.3	77.2	89.3	2.828e-159
rfsrc_lr	aorsf_lcpc	-0.0	-0.0	-0.0	1.955e-03
rfsrc_lrsc	aorsf_lcpc	-0.0	-0.0	-0.0	1.372e-04
rfsrc_bs	aorsf_lcpc	-0.0	-0.0	-0.0	1.988e-04
cox	aorsf_lcpc	83.2	77.2	89.3	3.912e-159
rfsrc_lrsc	rfsrc_lr	-0.0	-0.0	-0.0	4.071e-03
rfsrc_bs	rfsrc_lr	-0.0	-0.0	-0.0	4.907e-03
cox	rfsrc_lr	83.3	77.2	89.3	3.082e-159
rfsrc_bs	rfsrc_lrsc	0.0	-0.0	0.0	2.716e-01
cox	rfsrc_lrsc	83.3	77.2	89.3	2.918e-159
cox	rfsrc_bs	83.3	77.2	89.3	2.966e-159

Table A5. Significance of Mean Differences Comparison T-tests for Brier Score values obtained by 10-fold (3 years).

Model	Reference	Delta.Brier	Lower	Upper	p
aorsf	Null model	-0.3	-0.4	-0.2	1.386e-29
aorsf_lc	Null model	-0.3	-0.4	-0.2	1.247e-29
aorsf_lcpc	Null model	-0.2	-0.2	-0.1	2.348e-07
rfsrc_lr	Null model	-0.3	-0.3	-0.2	1.533e-15
rfsrc_lrsc	Null model	-0.3	-0.4	-0.2	9.834e-18
rfsrc_bs	Null model	-0.3	-0.4	-0.2	3.059e-17
cox	Null model	83.0	76.9	89.1	3.189e-158
aorsf_lc	aorsf	0.0	-0.0	0.0	6.086e-01
aorsf_lcpc	aorsf	0.1	0.1	0.2	1.533e-09
rfsrc_lr	aorsf	0.0	0.0	0.1	3.672e-02
rfsrc_lrsc	aorsf	0.0	-0.0	0.0	5.338e-01
rfsrc_bs	aorsf	0.0	-0.0	0.0	2.939e-01
cox	aorsf	83.3	77.2	89.4	1.458e-159
aorsf_lcpc	aorsf_lc	0.1	0.1	0.2	1.855e-09
rfsrc_lr	aorsf_lc	0.0	0.0	0.1	4.296e-02
rfsrc_lrsc	aorsf_lc	0.0	0.0	0.0	5.691e-01
rfsrc_bs	aorsf_lc	0.0	0.0	0.0	3.217e-01
cox	aorsf_lc	83.3	77.2	89.4	1.472e-159
rfsrc_lr	aorsf_lcpc	-0.1	-0.2	-0.1	2.727e-05
rfsrc_lrsc	aorsf_lcpc	-0.1	-0.2	-0.1	7.162e-07
rfsrc_bs	aorsf_lcpc	-0.1	-0.2	-0.1	1.580e-06
cox	aorsf_lcpc	83.2	77.1	89.2	4.621e-159
rfsrc_lrsc	rfsrc_lr	-0.0	-0.0	-0.0	6.842e-04
rfsrc_bs	rfsrc_lr	-0.0	-0.0	-0.0	1.205e-03
cox	rfsrc_lr	83.3	77.3	89.4	5.510e-160
rfsrc_bs	rfsrc_lrsc	0.0	-0.0	0.0	2.893e-01
cox	rfsrc_lrsc	83.3	77.3	89.4	4.776e-160
cox	rfsrc_bs	83.3	77.3	89.4	4.857e-160

Table A6. Significance of Mean Differences Comparison T-tests for Brier Score values obtained by 10-fold (5 years).

Model	**Reference**	**Delta.Brier**	**Lower**	**Upper**	**p**
aorsf	Null model	-0.7	-0.8	-0.6	1.422e-56
aorsf_lc	Null model	-0.7	-0.8	-0.6	8.946e-57
aorsf_lcpc	Null model	0.1	-0.0	0.3	6.335e-02
rfsrc_lr	Null model	-0.7	-0.8	-0.6	1.00e-34
rfsrc_lrsc	Null model	-0.7	-0.8	-0.6	1.465e-39
rfsrc_bs	Null model	-0.7	-0.8	-0.6	1.273e-37
cox	Null model	82.6	76.6	88.7	8.459e-157
aorsf_lc	aorsf	0.0	-0.0	0.0	6.807e-01
aorsf_lcpc	aorsf	0.0	0.7	1.0	1.011e-42
rfsrc_lr	aorsf	0.0	0.0	0.1	2.707e-02
rfsrc_lrsc	aorsf	0.0	-0.0	0.1	2.756e-01
rfsrc_bs	aorsf	0.0	-0.0	0.1	1.432e-01
cox	aorsf	83.4	77.3	89.4	5.121e-160